

# ADVAITH RENJITH

Rutgers University | Mechanical & Aerospace Engineering | Class of 2030

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Research interests: fluid dynamics and physics-informed modeling; autonomous systems; soft robotics.

## EDUCATION

**Rutgers University, School of Engineering** | B.S. in MAE, expected May 2030

**Edison Academy Magnet School** | Class of 2026

SAT 1580 (Math 800); AP Physics C Mechanics / E&M: 5 / 5; AP Calculus BC: 5.

## RESEARCH & ENGINEERING

**QuGTrans: Physics-Informed Flood Forecasting** | 2025-2026

*2nd Place, NJ State Science Fair - Computer Science Division*

- Achieved 117× faster inference than a finite-difference reference in project benchmarks, with 97% mass-conservation adherence.
- Added the 1D Saint-Venant continuity equation to the training loss to constrain predicted flood dynamics by conservation of mass, combining fast learned forecasts with a physical consistency objective.

**Soft Robotics Arm** | Independent project, 2024-2026

- Built a three-cable, segmented thermoplastic polyurethane (TPU) continuum arm; used differential cable tension and material compliance for spatial actuation.
- Integrated vision-language guidance and heuristic control with pose-driven actuation for 3D trajectory tracking.

**Active Aerodynamics Testbed** | Independent project, 2023-2026

- Built a 1/8-scale RC platform with a ground-effect downforce fan and drag-reduction system for active-aerodynamics experiments.
- Repurposed a smartphone as the vehicle controller and telemetry unit for on-device sensing and test-data collection.

**NASA App Development Challenge - Team Luna Vista** | 2025

*National Winner, 200+ teams*

- Modeled the Artemis II flight path in Unreal Engine 5 using orbital mechanics and C++ shaders; optimized antenna scheduling for the winning visualization.

## PUBLICATION

**Beyond the Haystack: Sensitivity to Context in Legal Reference Recall**

Third author | NLLP Workshop at EMNLP 2025 | Accepted 2025

- Designed an evaluation framework for long-context recall across three naturalistic legal-reference tasks using U.S. court-opinion briefings; exposed retrieval behavior missed by synthetic benchmarks.

## TECHNICAL SKILLS

**Programming:** Python, C++ | **Modeling:** physics-informed learning, model evaluation, reinforcement learning

**Hardware:** TPU prototyping, cable actuation, RC telemetry, CubeSat design

**Tools:** Unreal Engine 5, CAD, on-device inference

## HONORS & SERVICE

**Sir Isaac Newton Physics Exam:** 9th globally among 3,800+ participants.

**USAPhO:** Distinction; top 400 nationally; F=MA qualifier.

**MIT Beaver Works Summer Institute:** 4th nationally; wildfire-detection CubeSat design.

**NJ Aviation Hall of Fame (2024-2026):** digitized 8,000+ pages of aerospace archives.

**InfoAge Science & History Museums (2024-2026):** artifact database spanning 20 buildings; technical repairs and public interpretation.